

Integrated Sensing and Communication in Agentic LAWNs: Challenges, Frameworks, and Opportunities

Wenwen Xie, Geng Sun, Jiahui Li, and Zemin Sun

ABSTRACT

Low-altitude wireless networks (LAWNs) have emerged as a crucial paradigm for supporting a wide range of low-altitude applications. However, dynamic environments and tightly coupled functions require more intelligent and autonomous operation. In this case, agentic artificial intelligence (AI) offers a viable solution by enabling a continuous perception-reasoning-action loop within dynamic environments. Meanwhile, integrated sensing and communication (ISAC) enables simultaneous sensing and communication within a unified wireless framework, thereby supporting intelligent and context-aware services for LAWNs. As such, we investigate the application value and development prospects of ISAC in agentic LAWNs in this work. First, we provide a comprehensive discussion on the relationship between agentic LAWNs and ISAC to demonstrate their complementary characteristics in supporting low-altitude wireless applications. Second, we expand the perspective of ISAC in LAWNs by presenting the integration of semantic communication, computation, and control, particularly highlighting the significant advantages of agentic AI in coordinating multiple system functions. Third, we propose a novel agentic AI framework for ISAC in LAWNs and present a case study to verify its superiority in optimizing sensing and communication performance. Finally, we clarify future research directions.

INTRODUCTION

With the rapid expansion of low-altitude applications, wireless networks operating in the low-altitude space have attracted increasing attention in both academia and industry [1]. Typical applications include aerial surveillance, emergency response, environmental monitoring, cargo delivery, and intelligent transportation. In this case, low-altitude wireless networks (LAWNs) have emerged as a promising networking paradigm [2]. By deploying aerial platforms such as unmanned aerial vehicles (UAVs), LAWNs can provide on-demand wireless coverage and flexible service

provisioning. Compared with conventional terrestrial networks, LAWNs can rapidly adjust network topology through node mobility and can establish line-of-sight (LoS) links more easily. These advantages enable LAWNs to support diverse wireless services in complex environments and make them an important component of future sixth-generation (6G) wireless systems.

Despite the potential of LAWNs, improving the performance of such networks remains a challenging task. Existing studies often rely on traditional optimization methods to design resource allocation and trajectory control strategies. Although these methods provide theoretically optimal or near-optimal solutions under specific assumptions, they usually require accurate mathematical models and complete environmental information. However, the wireless environment in LAWNs is highly dynamic due to node mobility, time-varying channels, and diverse service requirements. Under these conditions, traditional optimization approaches often suffer from high computational complexity and limited adaptability. As such, it becomes difficult for these methods to provide efficient decision-making in large-scale and dynamic LAWNs.

Recently, agentic artificial intelligence (AI) has emerged as a promising paradigm for intelligent decision-making in complex systems [3]. Different from conventional AI methods that focus mainly on perception or prediction tasks, agentic AI emphasizes autonomous decision-making through a perception-reasoning-action loop. In this paradigm, agents can automatically decompose complex tasks into simpler sub-tasks, perceive the environment, analyze information through reasoning mechanisms, and proactively initiate actions to achieve specific goals. Meanwhile, the agents can store historical experiences in memory and continuously improve their decision strategies through interaction with the environment. These characteristics enable agentic AI to adapt to dynamic environments and support complex tasks that require long-horizon planning [4]. Therefore, agentic AI provides a potential solution

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to address the challenges of decision-making and resource management in LAWNs and forms agentic LAWNs.

Moreover, integrated sensing and communication (ISAC) plays an important role in LAWNs [5] since it enables unified sensing and communication to improve spectrum efficiency and supports simultaneous data transmission and environmental perception for applications such as aerial surveillance, autonomous navigation, and target tracking. Meanwhile, agentic LAWNs provide additional advantages for ISAC deployment because the high mobility and flexible positioning of aerial platforms can dynamically improve sensing perspectives and communication links beyond fixed terrestrial base stations, while agentic AI can effectively address the coupling between trajectory control and resource allocation by enabling autonomous coordination of perception and communication in dynamic environments. Therefore, there is a complementary relationship between ISAC and agentic AI.

Accordingly, we explore the ISAC in agentic LAWNs in this work. The main contributions of this work are summarized as follows

- We first present a comprehensive discussion on the relationship between agentic LAWNs and ISAC. In particular, we analyze the advantages of LAWNs in supporting low-altitude wireless applications and highlight how agentic AI can further enhance the intelligence and adaptability of these networks. Moreover, we further clarify the complementary characteristics between agentic LAWNs and ISAC.
- We further expand the perspective of ISAC in LAWNs. Specifically, we discuss the application of semantic communication, computation, and control integration in ISAC within LAWNs, highlighting how agentic AI can coordinate multiple system functions and provide smarter, more efficient wireless services for complex low-altitude application scenarios.
- We propose an agentic AI framework for ISAC in agentic LAWNs that integrates perception, reasoning, memory, and action into a unified decision loop. Moreover, we develop a case study to validate the effectiveness of the proposed framework. Simulation results demonstrate that the proposed framework significantly improves performance and exhibits strong adaptability to dynamic wireless environments.

OVERVIEW OF AGENTIC LAWN AND ISAC

In this section, we provide a detailed overview of the agentic LAWNs and ISAC.

DEFINITION AND ANALYSIS OF AGENTIC LAWN

Characteristics and Challenges of LAWN: The rapid proliferation of UAVs and the increasing demand for low-altitude economy applications have catalyzed the emergence of LAWNs. Specifically, LAWNs typically leverage aerial platforms operating within the low-altitude airspace to provide diverse and flexible connectivity for ground nodes, which represents a transformative paradigm shift in networking. Compared with conventional terrestrial networks, LAWNs provide wider and more flexible wireless coverage, enhanced reliability, and stronger adaptability to dynamic

environments through the mobility and cooperation of aerial platforms [2].

Nevertheless, these advantages make performance optimization much harder for conventional optimization methods. Specifically, they rely on static or slowly varying models and thus cannot respond in real time to highly dynamic environments. Moreover, they are designed for specific low-level tasks, whereas LAWNs involve high-level complex problems that require manual decomposition or expert heuristics, increasing design complexity and reducing generality. Given the aforementioned limitations, developing a new optimization paradigm for LAWNs has become urgent. In this case, agentic LAWNs that integrate agentic AI have attracted unprecedented attention and are regarded as a promising research direction.

Concept of Agentic AI and Agentic LAWN:

The development of AI has generally progressed through three representative stages. First, the introduction of the Q-learning algorithm in 1989 promoted the rapid advancement of reinforcement learning (RL), while the emergence of deep Q-network (DQN) in 2015 further expanded the application of deep reinforcement learning (DRL) to high-dimensional decision-making problems. However, RL involves a black-box decision-making process and lacks the capability for general knowledge-based understanding and reasoning. Consequently, the large language model (LLM) gradually emerged, where the Transformer architecture proposed in 2017 provided the technological foundation. Moreover, the release of GPT-3 in 2020 accelerated the development of LLMs, thereby giving rise to representative applications such as ChatGPT. Nevertheless, LLMs essentially remain passive response systems because their behavior depends on user instructions. To achieve a higher level of autonomous decision-making and execution capability, agentic AI emerged as a new research direction and produced representative frameworks such as AutoGen in 2023. Therefore, agentic AI has become an important research focus in the field of AI [6].

Specifically, agentic AI can autonomously decompose complex high-level tasks into low-level sub-tasks, and then assign these sub-tasks to agents integrating different algorithms and external tools to process them based on perceived external environmental state and historical knowledge, thereby generating and executing the final decision [7]. As we can see, agentic AI follows a perception-reasoning-action loop and exhibits the following characteristics.

- *Autonomy.* Unlike conventional AI methods that often require substantial human involvement, agentic AI can perform reasoning and make decisions with minimal human intervention, enabling autonomous task execution toward predefined objectives.
- *Explicit Reasoning.* Conventional AI methods often operate as black boxes, whereas agentic AI employs explicit reasoning mechanisms, thereby improving interpretability and reliability.
- *Task Decomposition and Multi-agent Collaboration.* Agentic AI can decompose complex tasks into manageable sub-tasks and coordinate multiple specialized agents, thereby improving processing efficiency and decision quality.
- *Memory Capability.* Agentic AI can leverage

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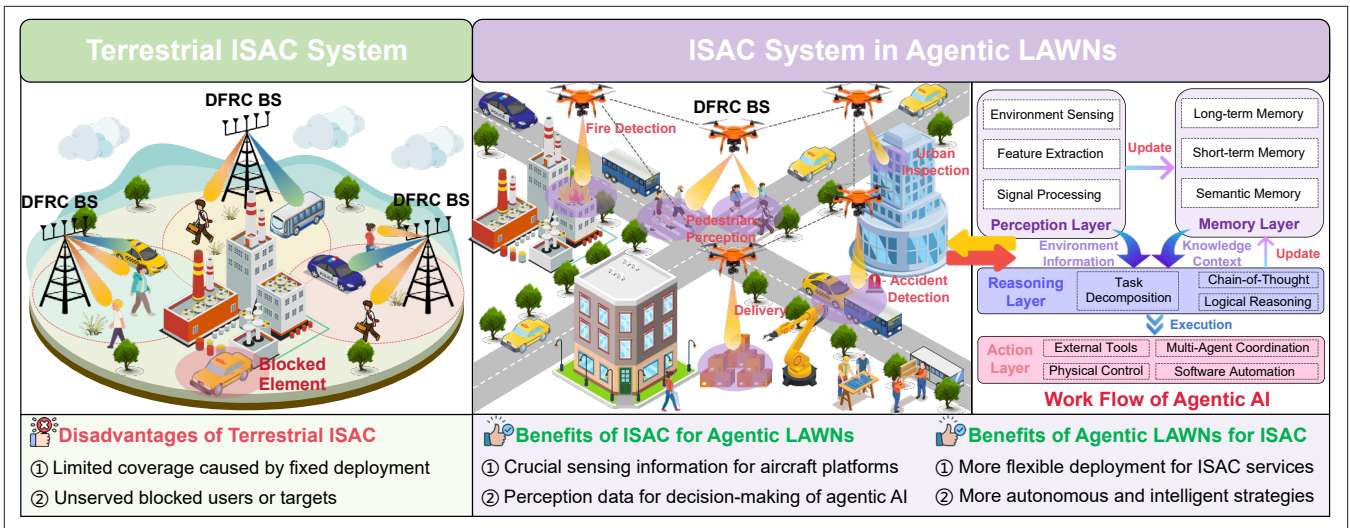


FIGURE 1. Comparisons of terrestrial ISAC systems and ISAC systems in agentic LAWNs.

historical decisions and feedback to support future reasoning, thereby continuously improving decision-making performance [8].

Agentic LAWNs refer to LAWNs that incorporate agentic AI into the core processes of network management, control, and optimization, enabling intelligent agents to actively participate in the network operation and decision-making process [9]. Compared with conventional LAWNs, agentic LAWNs provide more adaptive decision-making in dynamic environments by reducing reliance on handcrafted rules and empirical models. Moreover, multi-agent collaboration enables efficient coordination across different network functions, thereby improving operational efficiency and overall network performance.

COMPLEMENTARITY BETWEEN ISAC AND AGENTIC LAWNS

ISAC refers to a cyber-physical system with integrated sensing and communication capabilities, enabling real-time environmental monitoring and data-driven intelligent control. Based on hardware deployment related to sensing and communication functionalities, ISAC includes the following two architectures, namely radar-communication coexistence (RCC) and dual-functional radar-communication (DFRC) [10]. Compared to the RCC architecture, the DFRC architecture achieves tighter coupling between sensing and communication functionalities, improving spectrum efficiency while enabling more efficient resource management in dynamic scenarios, which makes it a more popular and promising mainstream ISAC architecture in agentic LAWNs.

Despite its advantages, the DFRC architecture still faces several critical bottlenecks in further performance improvements, as shown in Fig. 1. First, most existing DFRC systems are deployed on terrestrial infrastructures, which makes their performance susceptible to obstacle blockage, complex terrain, and limited coverage. Second, DFRC architectures impose more stringent requirements on signal design and resource allocation, since sensing and communication functionalities should be jointly realized using unified waveforms and shared spectrum resources. In this case, agentic LAWNs provide an effective way to address the aforementioned challenges faced by the DFRC architec-

ture. Specifically, the low-altitude aerial platforms (e.g., UAVs) are deployed as aerial base stations (BSs) to simultaneously perform sensing and communication tasks, which overcomes the blockage and coverage limitations faced by terrestrial infrastructure due to the high mobility of aerial nodes [11]. Moreover, agentic AI supports continuous interaction with the environment and actively analyzes the trade-off between sensing and communication tasks through explicit reasoning mechanisms, thereby generating and iteratively refining beamforming design and resource allocation strategies in a step-by-step manner, rather than relying solely on passive policy fitting based on conventional data-driven approaches [12].

BEYOND ISAC: EXTENDED AGENTIC LAWN PARADIGMS

In this section, we broaden the discussion from conventional ISAC to several emerging sensing-communication-computing paradigms. Specifically, integrated sensing and semantic communication (ISSC) focuses on both sensing and semantic communication functions to improve the efficiency of perception information delivery. Building upon ISSC, integrated sensing, communication, and computing (ISCC) further introduces computation capabilities through edge intelligence, providing the necessary computational support for complex AI inference and environmental modeling in LAWN. Finally, integrated sensing, communication, computing, and control (ISC3) extends ISCC by incorporating control functionalities, thereby establishing a complete cyber-physical closed loop. In the following, we introduce the specific applications of the different extended versions of ISAC in agentic LAWNs.

INTEGRATED SENSING AND SEMANTIC COMMUNICATION

In continuous low-altitude sensing scenarios, airborne nodes should capture and transmit high-dimensional environmental data under stringent latency constraints. However, LAWN faces severe energy constraints and dynamic topologies, which limits the effectiveness of traditional transmission modes for high-data-volume tasks. In this case, ISSC effectively addresses these critical bottlenecks by extracting and transmitting task-relevant semantic information rather than transmitting pre-

cise data. As such, this technology significantly reduces onboard power consumption for wireless transmission and minimizes latency, which ensures continuous mission endurance for airborne nodes even in resource-constrained environments. However, ISCC typically involves complex decision variables due to the semantic compression process, making it difficult for single, conventional algorithms to make appropriate decisions directly. In this case, agentic AI is a viable approach due to its ability to decompose complex problems.

For example, the authors in [13] investigated ISCC in agentic LAWNs. Specifically, they considered a semantic wireless network, where a UAV performs the target sensing tasks and then transmits the semantic sensing information to a mobile BS. To improve operational endurance, the authors proposed an agentic AI framework that can be considered as an intelligent orchestration pipeline designed to systematically decouple and resolve the highly coupled variables. Specifically, the framework first utilizes the positioning agent to determine the optimal spatial deployment of the aerial node by geometrically evaluating the sensing performance constraints. Following the spatial configuration, the framework activates the compression agent to dynamically calibrate the cognitive processing load by analytically deriving the optimal semantic compression ratio. Finally, the framework employs the offloading agent to perform a deterministic search over feasible transmit powers, and then compares the energy efficiency of localized large AI model inference at the edge versus offloaded computation at the cloud server. Simulation results demonstrated that this agentic AI framework effectively ensures global optimality and significantly reduces total energy consumption compared to conventional baselines.

INTEGRATED SENSING, COMMUNICATION, AND COMPUTATION

In agentic LAWNs, the inference process of agentic AI typically consumes significant computing power, which is challenging for UAVs. While deploying agentic AI to remote base stations and establishing backhaul links with UAVs is a common practice, this inevitably introduces significant backhaul latency. To address this issue, ISCC introduces edge computing capabilities into LAWNs, enabling edge servers to support the computational demands of agentic AI while facilitating overall network control. As such, this technology reduces the computational burden on airborne nodes and improves system adaptability under unstable backhaul conditions, which allows low-altitude wireless networks to respond rapidly to complex and dynamic scenarios.

For example, the authors in [14] considered an ISCC-enabled LAWN in which multiple UAVs communicate with multiple mobile users (MUs) based on their perceived environmental information and a base station acts as an edge node responsible for controlling and managing the entire network. Considering that the decision-making process is dependent on the sensing data, the authors proposed an agentic AI framework that separates environment modeling from resource allocation. Specifically, the framework first employs a perception agent implemented by a fine-tuned multi-modal large language model (M2LLM) to process heterogeneous sensing inputs, including images, wireless measurements,

and textual task descriptions, and then predicts MU mobility and extracts a compact environment state representation. Subsequently, the framework schedules a DRL-enabled agent to analyze the extracted representation to learn adaptive strategies for trajectory control and beamforming design. Through this hierarchical workflow, perception intelligence provided by multimodal reasoning improves environment awareness, while the DRL performs sequential control optimization based on the predicted system evolution. Simulation results demonstrated the effectiveness of the proposed agentic AI framework.

INTEGRATED SENSING, COMMUNICATION, AND CONTROL

A key bottleneck in LAWNs with integrated ISCC lies in the fundamental ability to translate computational information into immediate physical actions. This can be addressed by integrating control functions into the existing framework to form an ISC3 system. Specifically, ISC3 establishes a closed-loop cyber-physical architecture, where sensing modules capture environmental and network states, communication links enable state exchange among distributed nodes, computation units located at base stations or control centers execute network-level algorithms for decision generation, and control modules translate these decisions into physical actuation signals that drive system dynamics. Moreover, given that the ISC3 system involves tightly coupled functions, agentic AI offers a promising approach by adopting a multi-agent mechanism that facilitates collaboration and adaptive interaction among agents responsible for different system functions.

For example, the authors in [15] proposed an embodied agentic AI framework to autonomously orchestrate ISC3 functionalities. Specifically, the framework first employs a cognition agent to interpret heterogeneous application demands and translate them into explicit operational objectives. Subsequently, a perception agent aggregates and fuses multimodal data from onboard sensors to construct a comprehensive environmental representation. Following this, the framework activates a decision agent that leverages the reasoning capabilities of the M2LLM to dynamically synthesize precise spatial navigation strategies and wireless resource allocation schemes. Finally, a control agent translates the decisions into physical flight maneuvers. To verify the performance of this framework, the authors considered a UAV express delivery scenario and minimized the flight route length. The simulation results demonstrated that the proposed framework can efficiently process multimodal data to optimize flight paths and communication links.

CASE STUDY: AGENTIC LAWNS FOR ISAC

In this section, we propose an agentic framework for ISAC in LAWNs and construct a case study to evaluate its performance.

SCENARIO AND MOTIVATIONS

We consider a UAV-enabled ISAC system, where a UAV with multiple antennas acts as an aerial BS to perform communication tasks with multiple terrestrial users while sensing a target. To improve communication capacity and sensing accuracy, we formulate a multi-objective optimization problem to maximize the communication rate with

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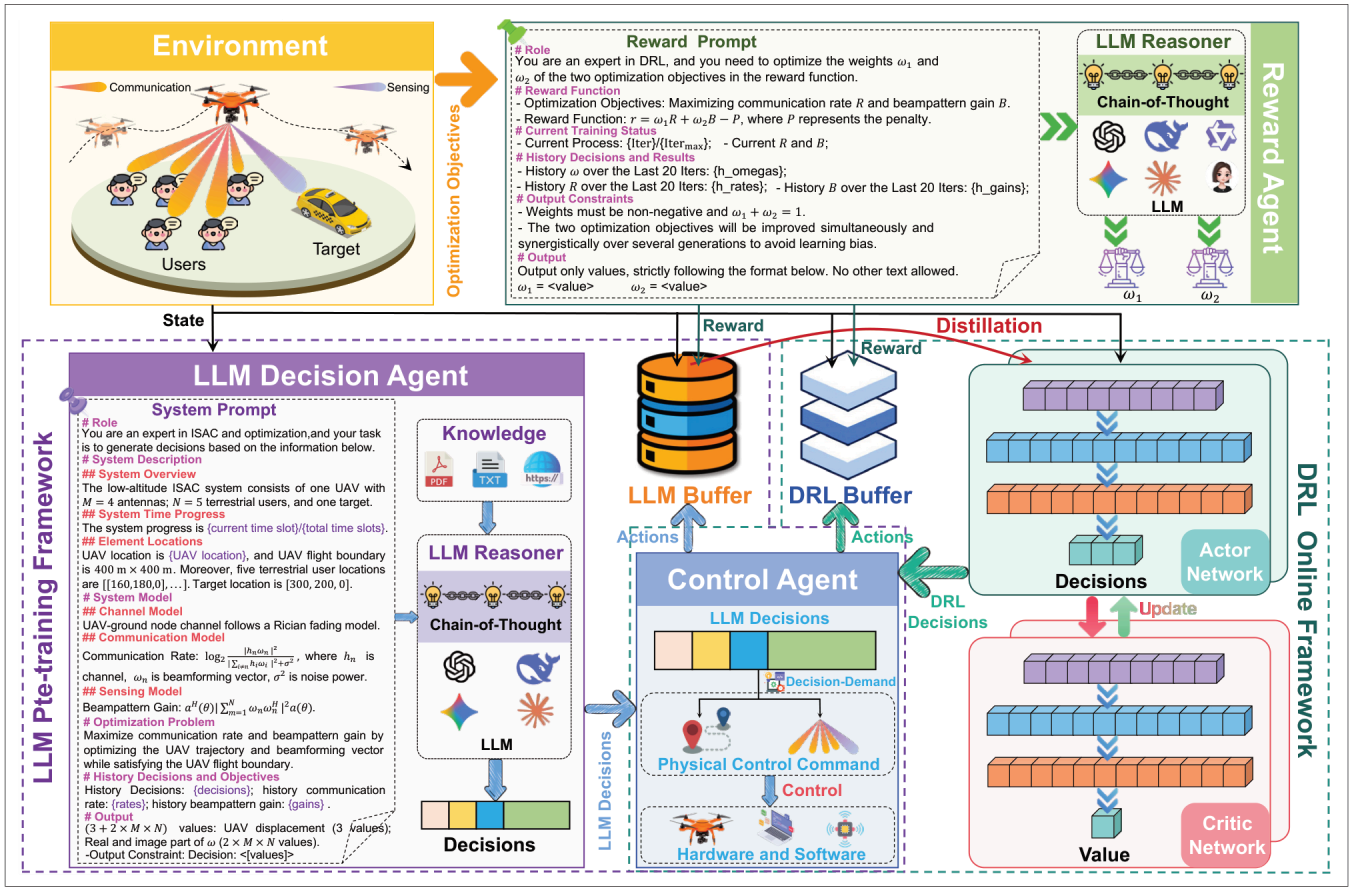


FIGURE 2. Agentic AI Framework for ISAC in LAWNs. Specifically, the LLM decision agent first generates an expert strategy, which is distilled into the DRL decision agent to provide high-quality initialization. Subsequently, the DRL decision agent refines its policy using the feedback from the reward agent. Finally, the trained DRL policy is deployed for real-time decision making.

users and the target beampattern gain by optimizing the UAV trajectory and beamforming. To solve the challenging optimization problem, we propose an agentic AI framework.

AGENTIC AI-BASED FRAMEWORK

Framework Design: As shown in Fig. 2, the proposed framework consists of several collaborative agents that perform different functions within the decision process.

- **LLM Decision Agent:** The LLM decision agent acts as a high-level expert that provides reference decisions. This is because LLMs contain extensive general knowledge and strong reasoning capability, which enables them to generate informed decisions. Meanwhile, the LLM decision agent also updates its decision strategy based on feedback from previous decisions, which gradually improves the decision quality over time.
- **DRL Decision Agent:** The DRL decision agent focuses on efficient online decision-making. Although the LLM decision agent can provide high-quality reasoning, the inference process typically requires considerable computational resources and time. This requirement is challenging for airborne nodes such as UAVs with limited onboard computing capability. In contrast, DRL algorithms are relatively lightweight and support fast real-time inference, making them more suitable for real-time decision-making based on observed system states.
- **Reward Agent:** The reward agent evaluates

the quality of decisions generated by both the LLM decision agent and the DRL decision agent. In ISAC systems, communication performance and sensing performance typically exhibit a trade-off relationship. In this case, the LLM-based reward agent dynamically adjusts the weights between communication and sensing objectives according to the current system performance, which improves the reliability of the evaluation signal and facilitates more balanced optimization.

- **Control Agent:** The control agent converts the generated decisions into executable control commands based on interface encapsulation. Specifically, after receiving the outputs from the decision agents, the control agent converts them into specific electrical signals or actuator commands through the underlying hardware driver interface. This design improves system reusability because the decision agents do not require retraining when the underlying physical platform changes, and only the corresponding interface needs to be modified. These converted commands are then applied to the physical devices or software modules in the system, which enables practical implementation of the decision results.

The proposed agentic AI framework follows a perception-reasoning-action loop, and its detailed operational process is as follows. First, the system perceives the environment state and constructs a professional system prompt that contains the system model, optimization objective, and historical

decision information. This information is provided to the LLM decision agent, which performs step-by-step reasoning and interacts with the environment iteratively to improve its strategy through feedback from the reward agent. After sufficient interactions, the LLM decision agent learns a refined decision strategy. The learned strategy is then distilled into the DRL decision agent to facilitate subsequent learning. Guided by the distilled knowledge, the DRL decision agent begins its training process and continues to improve its strategy using the reward signals provided by the reward agent. After training, the trained DRL agent generates real-time decisions based on the observed states. Finally, the control agent converts the DRL decision into executable control commands and deploys them in the system. As we can see, the proposed framework does not require the development of a new underlying architecture. Instead, it adopts an integration strategy based on mature technologies, which ensures a solid foundation for practical deployment within current software and hardware ecosystems.

Feasibility Analysis of the Framework: First, the proposed agentic AI framework adopts asymmetric computation deployment and policy distillation to reduce real-time computation overhead. Specifically, LLM reasoning and DRL training are executed on edge servers or ground stations, while UAVs only run a distilled lightweight DRL policy. Although this design introduces additional offline training and distillation costs, it preserves the reasoning benefits of LLMs while significantly reducing deployment complexity compared to the online LLM deployment. Second, although the quality of sensing data cannot always be guaranteed in dynamic environments, the proposed agentic AI framework exhibits stronger robustness than conventional approaches. Specifically, conventional methods heavily depend on accurate models and real-time state information, whereas agentic AI can combine historical interaction information and LLM-based knowledge for assisted reasoning, thus generally exhibiting better adaptability when data quality degrades. In other words, although the performance of agentic AI remains influenced by sensing quality, its underlying mechanism can partially mitigate the impact of data uncertainty. Finally, the proposed agentic AI framework provides strong scalability. For large-scale LAWN scenarios, the DRL decision agent can be extended to a multi-agent DRL architecture with centralized training and distributed execution. Therefore, larger-scale networks can be supported without modifying the overall framework.

SIMULATION RESULTS

In the simulation, we select the DeepSeek model because of its advantages in inference cost and response speed. Notably, since the proposed framework is independent of specific LLMs, it can integrate other open-source and closed-source LLMs according to the task-specific requirements for privacy, response speed, and computing power.

Figure 3 shows the achieved optimization objective values under different transmit power settings. As can be seen, the proposed agentic AI framework consistently outperforms SAC. This performance occurs because the LLM decision agent provides high-quality expert initialization based on

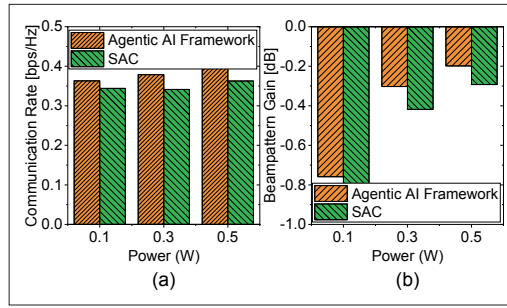


FIGURE 3. Optimization objectives obtained by different algorithms: a) communication Rate; b) beampattern Gain.

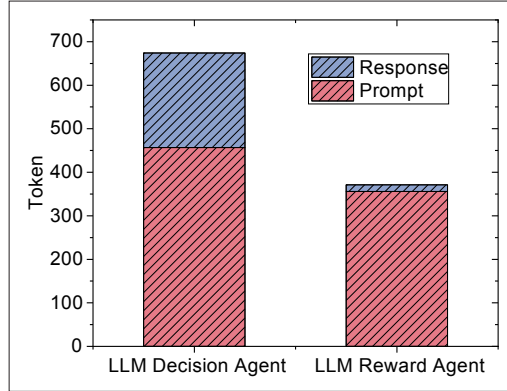


FIGURE 4. Number of tokens used by different LLM-related agents per time.

general knowledge and reasoning capabilities, and the LLM reward agent continuously guides policy optimization toward more desirable directions. Consequently, the obtained policy achieves a superior communication rate and beampattern gain compared with the SAC algorithm that relies solely on exploration. These results demonstrate the effectiveness of LLM reasoning in providing meaningful guidance for high-dimensional and non-convex optimization problems.

Figure 4 presents the token consumption of the prompt and response components for the two LLM-related agents during a complete decision cycle. As can be seen, the total token consumption of the LLM decision agent is higher because it is responsible for processing system descriptions, decomposing tasks, and generating expert policy recommendations. By comparison, the LLM reward agent consumes 371 tokens in total, while only 15 tokens are required for its response generation. This is because this agent mainly provides concise reward adjustment suggestions. Notably, the UAV executes only the distilled lightweight DRL model during online operation, which indicates that no token consumption is introduced during real-time deployment.

FUTURE DIRECTIONS

Lightweight Agentic AI for Resource-Constrained LAWNS: In LAWNS, UAVs and edge nodes typically operate under stringent constraints in terms of onboard computation capability, memory capacity, and energy source. Therefore, developing lightweight agentic AI architectures is essential for practical deployment in LAWNS. Future research may investigate model compression, knowledge distillation, and parameter-efficient fine-tuning techniques to reduce the

This information is provided to the LLM decision agent, which performs step-by-step reasoning and interacts with the environment iteratively to improve its strategy through feedback from the reward agent.

complexity of agentic AI while maintaining reliable decision-making capability.

Secure and Trustworthy Agentic AI for ISAC: Agentic AI-enabled ISAC systems rely on shared information and environmental observations for autonomous decision making. Therefore, malicious data manipulation may degrade system reliability. To address this issue, blockchain and differential privacy can be incorporated to ensure data integrity and privacy protection, while adversarial training and anomaly detection can improve resilience against malicious attacks.

Knowledge-Augmented Reasoning for Complex ISAC Optimization: ISAC optimization in LAWNs involves highly coupled variables, making purely data-driven approaches insufficient in complex scenarios. Therefore, integrating domain knowledge into agentic AI is a promising direction. For example, knowledge graphs and physics-aware models can be combined with LLM-based reasoning to improve decision-making efficiency, interpretability, sample efficiency, and deployment reliability in large-scale LAWNs.

CONCLUSION

In this paper, we have investigated the ISAC in agentic LAWNs to support intelligent and autonomous wireless services. First, we have discussed the relationship between agentic LAWNs and ISAC and have analyzed their complementary characteristics in enabling diverse low-altitude applications. Second, we have extended the perspective of ISAC in LAWNs by discussing the integration of semantic communication, computation, and control, which demonstrates the potential of agentic AI in coordinating multiple system functions. Third, we have proposed an agentic AI framework for ISAC in LAWNs and have validated its effectiveness through a case study. Simulation results have shown that the proposed framework achieves superior sensing and communication performance compared with conventional approaches and demonstrates strong adaptability in dynamic wireless environments. Finally, we have outlined several promising research directions to facilitate the future development of agentic AI-enabled ISAC in LAWNs.

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